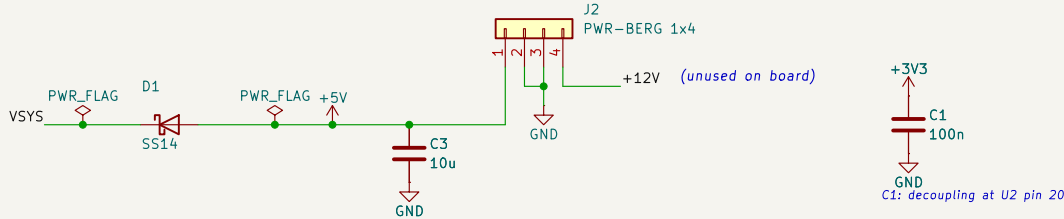


## POWER INPUT

Floppy 'Berg' power header. +5V feeds the module's VSYS through Schottky D1 so USB and floppy power can be connected together. +12V (J2 pin 4) is a header pad only: unused on the board. J2 is UNKEYED – reversed, +12V reaches VSYS.

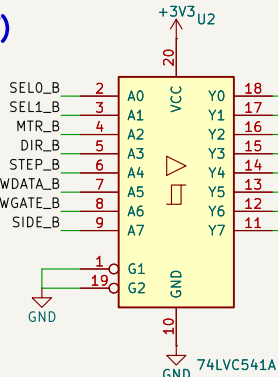
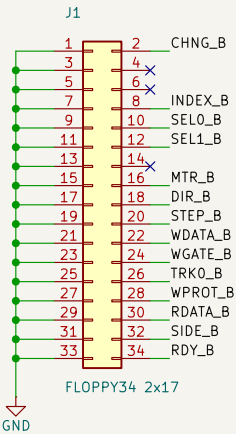


## U2 INPUT BUFFER Amiga → board

74LVC541A at 3V3: 5 V-tolerant inputs from the bus, 3.3 V outputs to the Pico. OE1/OE2 tied low (always on). Symbol body 74xx:74AHC541 (A0–A7/Y0–Y7 = A1–A8/Y1–Y8).

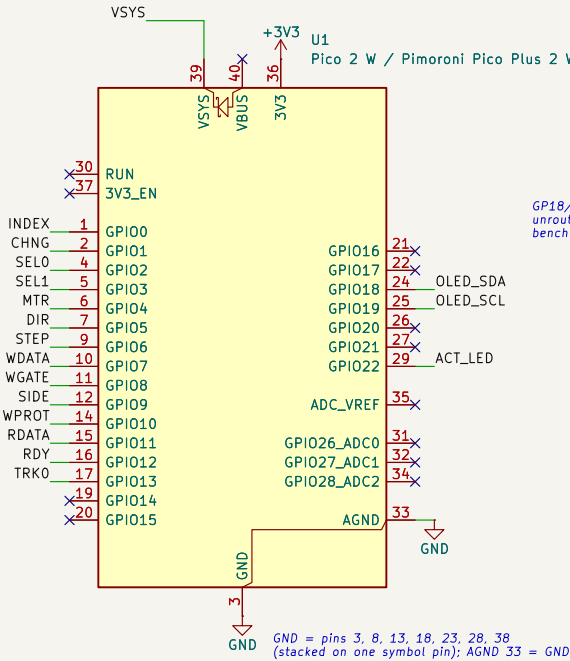
## J1 FLOPPY BUS (34-way, Amiga side, 5 V)

All odd pins GND. Pins 4, 6, 14 not connected. \*\_B nets = floppy-bus side of the buffers/drivers.



## U1 RP2350 MODULE (Pico 2 W / Pimoroni Pico Plus 2 W, PIM726)

Pin numbers = physical header pins 1–40. Net names = firmware names (floppy\_io.h). Powered from VSYS (pin 39); its 3V3 output (pin 36) supplies U2 and the OLED.

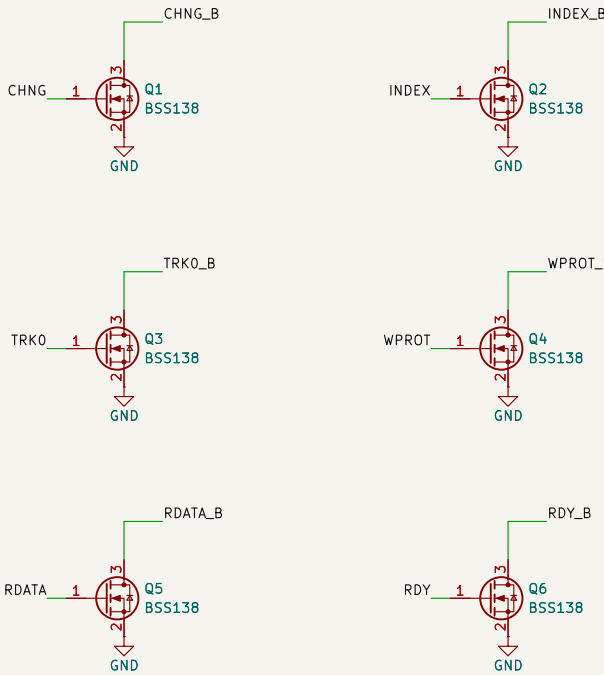


GP18/GP19/GP22 (pins 24/25/29): unrouted pads on the PCB – bench flying leads, see box

GND = pins 3, 8, 13, 18, 23, 28, 38 (stacked on one symbol pin); AGND 33 = GND

## Q1–Q6 OPEN-DRAIN DRIVERS board → Amiga

BSS138 N-FETs: GPIO high = bus line pulled low (inverting). No pull-ups on the board: the Amiga's own pull-ups set the high level.

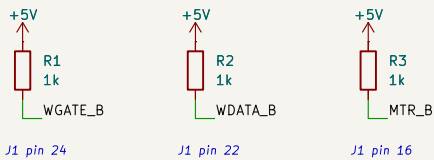


## BENCH MODIFICATIONS – hand-wired on rev A2, not on the PCB

Source: HANDOFF.md §4d (pull-ups, 2026–09–15) and the 2026–09–11 OLED / activity-LED notes.

### Floppy-line pull-ups, 1 kΩ to +5V (J2 pin 1)

On J1 pins 24 (WGATE), 22 (WDATA), 16 (MTR): host-driven lines the Amiga leaves open-collector. Without them WGATE floated; see HANDOFF 4d.

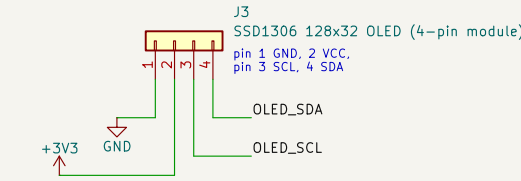


### Activity LED on GP22 (header pin 29)



!! No series resistor – see HANDOFF; add one on the next revision, (firmware limits the pad to 2 mA drive strength: a mitigation, not a fix)

### SSD1306 128x32 OLED module on I2C1



SDA = GP18 (pin 24), SCL = GP19 (pin 25), VCC = 3V3 (pin 36). GND = pin 23. The module's own pull-ups go to its VCC, so it must stay on 3V3; RP2350 GPIOs are not 5 V tolerant. Never use the PIM726 Qw/ST connector (hardwired to GP4/GP5 = MTR/DIR).

## NOTES

- Circuit = rev B PCB netlist (wifi\_floppy.kicad\_pcb). Rev A2, the board in hand, differs from rev B only in layout/silkscreen, not circuit.
- Amiga → board (SEL0, SEL1, MTR, DIR, STEP, WDATA, WGATE, SIDE): buffered by U2 (5 V-tolerant 74LVC541A).
- Board → Amiga (INDEX, CHNG, TRK0, WPROT, RDATA, RDY): open-drain via Q1–Q6: the high level comes from the Amiga's pull-ups.
- J1 odd pins 1–33 are all GND. J2: 1 = +5V, 2/3 = GND, 4 = +12V (unused).
- Connections are by net label: same name = same net.
- Bench parts (box, left) are hand-wired on rev A2, not on the PCB.

Source: hardware/schematic/generate\_schematic.py

Bench mods (R1–R3, D2, J3) are hand-wired on rev A2, NOT on the PCB  
Circuit: rev B PCB netlist = rev A2 as built (layout-only diffs) + bench mods

**Generated from wifi\_floppy.kicad\_pcb netlist + HANDOFF bench notes**

Sheet: /  
File: wifi\_floppy.kicad\_sch

**Title: WIFI FLOPPY – schematic (as built)**

Size: A3 Date: 2026–09–23

KiCad E.D.A. 10.0.4 Rev: A2 + bench mods Id: 1/1